

Altered postural control in PPPD: evidence for maladaptive mechanisms from motion capture posturography

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1. Introduction

Persistent Postural Perceptual Dizziness (PPPD) is a chronic functional vestibular syndrome with onset typically following an acute precipitating event e.g., vestibular neuritis¹.

Pathophysiological mechanisms of PPPD are not well established.

Force-plate posturography studies of functional dizziness disorders have identified transition to a stiffened postural control strategy².

We sought to assess postural control in PPPD via motion capture and pre-registered hypotheses that greater sway (vs. healthy controls) would diminish with more difficult stance and visual input conditions.

(AsPredicted # 250,803)

2. Methods

- Symptom characteristics assessed via Dizziness Handicap Inventory (DHI) and General Anxiety Disorder-7 (GAD-7) questionnaires.
- Quiet stance posturography in different stance (Romberg, tandem) and visual input (eyes open/closed) conditions (as shown in Fig. 1).
- Head kinematic time-series (20 s) recorded via marker-based motion capture, used to derive sway metrics, including sway area (95% prediction ellipse) from head trajectories (as shown in Fig. 2).
- Here we present an interim analysis.
- N = 33 (21 patients, PPPD, 12 healthy controls, HC).

Variable	HC (n=12)	PPPD (n=21)	P value
Age, mean (SD)	48.42 (14.93)	50.14 (11.63)	.734
Sex, n (%) Female	5 (41.7%)	15 (71.4%)	.092
DHI, median (IQR)	0 (0.0)	62 (18.0)	< .001
GAD-7, median (IQR)	2.0 (3.5)	3.0 (6.0)	.105

- Statistical comparison of log-transformed ellipse areas in a 2x(2x2) mixed ANOVA design.
- Spearman rank coefficient correlation of 95% prediction ellipse areas with DHI scores.

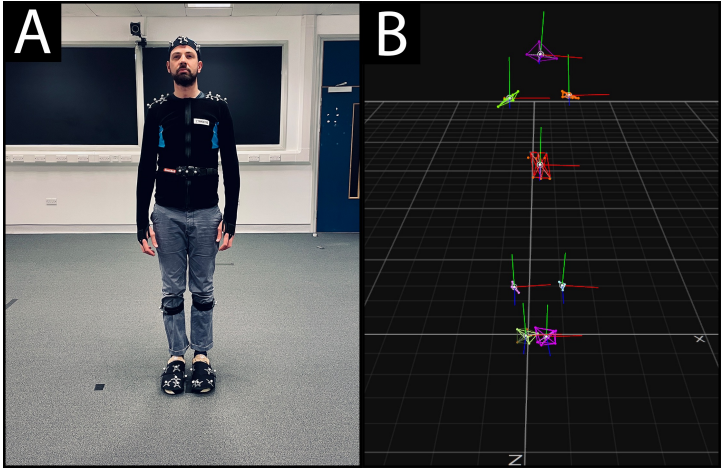


Figure 1. A: Marker-based motion capture setup. B: Markers reconstructed in Motive software for postural sway recording.

4. Discussion

- These interim findings provide evidence for altered postural control in PPPD consistent with maladaptive stiffening strategies, suggesting sensorimotor network dysfunction in the condition.
- The lower PPPD group sway in the most challenging condition may reflect 'locking down' of stance, with co-contraction of calf muscles as found before in EMG study of phobic postural vertigo³.
- Perceived dizziness-related disability was positively associated with objectively measured sway, which may be inconsistent with previous work emphasising postural misperception in PPPD⁴.
- Future work will integrate postural and visual-vestibular perceptual measures to investigate causal mechanisms underlying symptoms.
- Motion-capture posturography will be evaluated as a potential outcome measure for VR-based rehabilitation interventions.

STARDUST

The **ST**udy to **A**ssess and **R**educe **D**izziness Using **S**ensory **T**heory project is investigating:

- Optic flow parsing in PPPD.
- Multisensory integration (visual-vestibular mismatch).
- Walking performance in a VR-based obstacle avoidance task.
- Therapeutic approaches using manipulations of the VR visual environment.

References

- Popkirov et al. (2018). Practical Neurology, 18.
- Krafczyk et al. (1999). Neuroscience Letters, 259.
- Wuehr et al. (2017). Neurology, 88.
- San Pedro Murillo et al. (2023). JNNP, 94.

3. Results

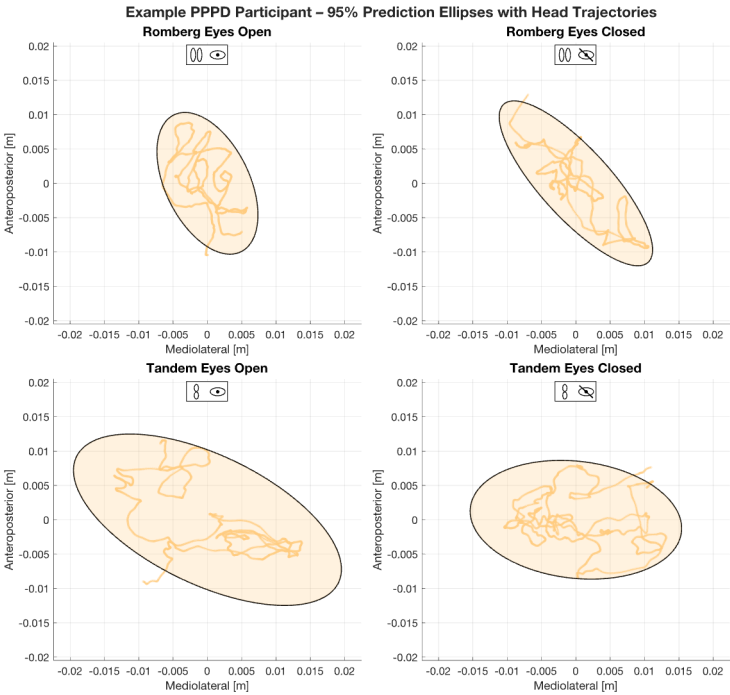


Figure 2. Head trajectories during quiet stance posturography with overlaid 95% prediction ellipses.

- Patients exhibited greater postural sway than controls, with sway increasing under more demanding balance conditions.
- However, during the most challenging condition, the PPPD group showed lower sway area than controls, potentially reflecting maladaptive postural stiffening (Fig 3.).

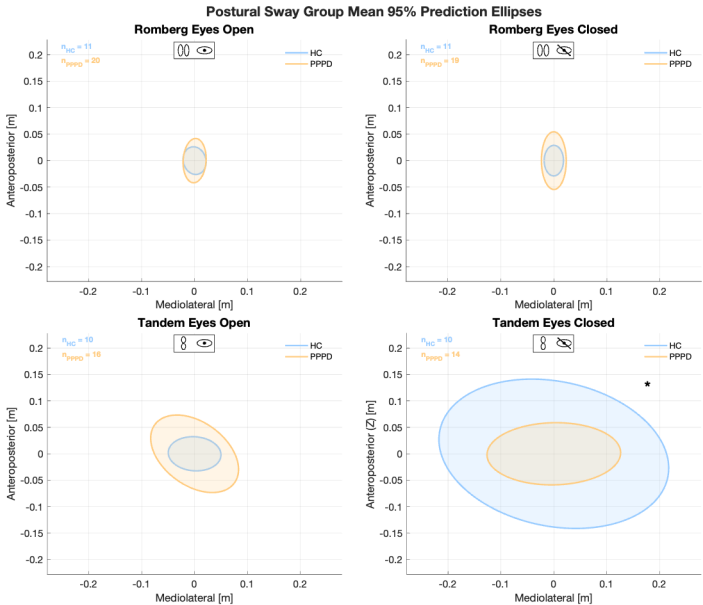


Figure 3. Mean 95% prediction ellipse area by group and condition.

*Condition contributing to the significant Group x Stance x Visual Input interaction, ($F(1,20) = 6.67, p = .018, \eta^2 = .25$), indicating that sway responses to sensory conditions differed between groups.

In the tandem eyes-closed condition, mean sway area was $0.066 \pm 0.036 \text{ m}^2$ in HC and $0.020 \pm 0.013 \text{ m}^2$ in PPPD.

- Dizziness-related disability (as measured by DHI) showed strong positive correlation with sway area in the PPPD group:

Romberg Eyes Open	$\rho = .71, p = .001 (n = 18)$
Romberg Eyes Closed	$\rho = .68, p = .002 (n = 17)$
Tandem Eyes Open	$\rho = .47, p = .076 (n = 15)$
Tandem Eyes Closed	$\rho = .57, p = .040 (n = 13)$